

date: 30-11-2023

Name ---

(12 marks)

Part A: Multiple choice

-Be sure that you have 6 different questions.

-Complete the following table by putting (the correct answer letter) in the table.

Q1	Q2	Q3	Q4	Q5	Q6
a	d	d	c	c	a

Q1) A class contain 4 math students, 5 physics and 6 chemistry students, suppose that 3 students are randomly selected, the probability that exactly 2 of the selected students are chemistry is

- a) 0.297 b) 0.45 c) 0.16 d) 0.25

Q2) If 5 dice are rolled, the probability that all show different faces is

- a) 0.36 b) 0.56 c) 0.4 d) 0.093

Q3) Let X be a random variable with mean 2 and standard deviation 3 then $E(2X^2 + X - 3) =$

- a) 26 b) 15 c) 40 d) 25

Q4) Three mobiles are drawn from 8 mobiles of which 5 are defective (معيوب), the probability that at least one is defective is

- a) 0.018 b) 0.008 c) 0.982 d) 0.792

Q5) In the experiment of the rolling two dice, what is the probability that the sum of results is 10.

- a) 2/15 b) 2/7 c) 1/12 d) 1/36

Q6) The following table provides the joint probability distribution of X and Y

$X \backslash Y$	1	2	3
1	0.10	0.30	0.1
2	0.1	0.2	0.2

The $P(X > 1 | Y = 3) =$

- a) 2/3 b) 2/5 c) 2/7 d) 2/9

Part B: Fill the Blanks

(8 marks)

Q1: An exam of applied statistics, noticed that 80% of student have no errors, 5% of students have one error and 15% have two errors, let x be random variable be number of errors, then the mean of errors will be 0.35

Q2: Let A and B be two independent events such that the probability is $1/6$ that they both occur and $1/3$ that A occurs and B does not occur. Then $P(B) = \frac{1}{3} \approx 0.33$

Q3: Two independent firefighting components A, B . Working such that $p(A)=0.2, p(B)=0.6$. Find the probability that both of them will fail. 0.88

Q4: let $f(x, y) = \begin{cases} e^x \sin y, & 0 \leq x \leq 2, 0 \leq y \leq \pi/2 \\ 0, & \text{otherwise} \end{cases}$

The joint cumulative distribution function is given by

$(e^x - 1)(1 - \cos y)$

Part C: Written Questions

(10 marks)

Question 1: Consider the following table that presents a random sample of 100 persons selected from the national institute of diabetes. One person is selected at random from the sample

	male	female
Diabetic	38	14
Non-diabetic	22	26

a) what is probability that the person is a male or the person has diabetes?

$$P(M \cup D) = P(M) + P(D) - P(M \cap D)$$

$$= \frac{60}{100} + \frac{52}{100} - \frac{38}{100} = \frac{74}{100}$$

Question 2: Let X be continuous random variable with Probability density function $f(x) = 2ax + 1$, $x \in [2, 4]$ find a .

$$\int_2^4 (2ax + 1) dx = 1$$

$$\left[\frac{2ax^2}{2} + x \right]_2^4 = 1$$

$$16a + 4 - [4a + 2] = 1$$

$$12a + 2 = 1 \Rightarrow 12a = -1 \Rightarrow a = -\frac{1}{12}$$

Question 3: A problem is given for two students S_1 and S_2 whose chances of solving it correctly are 0.5 and 0.75 respectively. Find the probability that the problem will be solved.

$$P(S_1 \cup S_2) = P(S_1) + P(S_2) - P(S_1 \cap S_2)$$

$$= 0.5 + 0.75 - [(0.5)(0.75)]$$

$$= 0.875$$

Best Wishes

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(12 marks)

Part A: Multiple choice

-Be sure that you have 6 different questions.

-Complete the following table by putting (the correct answer letter) in the table.

Q1	Q2	Q3	Q4	Q5	Q6
d	c	b	a	d	b

Q1) A class contain 4 math students, 5 physics and 6 chemistry students, suppose that 3 students are randomly selected, the probability that exactly 2 of the selected students are physics is

- a) 0.297 b) 0.45 c) 0.16 d) 0.198

Q2) If 6 dice are rolled, the probability that all show different faces is

- a) 0.36 b) 0.56 c) 0.015 d) 0.093

Q3) Let X be a random variable with mean 2 and standard deviation 3 then $E(2X^2 + X - 4) =$

- a) 26 b) 24 c) 40 d) 25

Q4) Three mobiles are drawn from 8 mobiles of which 5 are defective (معيوب), the probability that at least one is defective is

- a) 0.982 b) 0.008 c) 0.892 d) 0.792

Q5) In the experiment of the rolling two dice, what is the probability that the sum of results is 11.

- a) 2/15 b) 2/7 c) 1/12 d) 1/18

Q6) The following table provides the joint probability distribution of X and Y

X \ Y	1	2	3
1	0.10	0.30	0.1
2	0.1	0.2	0.2

The $P(X > 1 | Y = 2) =$

- a) 2/3 b) 2/5 c) 2/7 d) 2/9

Part B: Fill the Blanks

(8 marks)

Q1: An exam of applied statistics, noticed that 80% of student have no errors, 6% of students have one error and 15% have two errors, let x be random variable be number of errors, then the mean of errors will be 0.38

Q2: Let A and B be two independent events such that the probability is 1/6 that they both occur and 1/2 that A occurs and B does not occur. Then $P(B) = \frac{1}{4} = 0.25$

Q3: Two independent firefighting components A, B. Working such that $p(A)=0.2$, $p(B)=0.4$. Find the probability that both of them will fail. 0.72

Q4: let $f(x, y) = \begin{cases} e^x \sin y, & 0 \leq x \leq 2, 0 \leq y \leq \pi/2 \\ 0, & \text{otherwise} \end{cases}$

The joint cumulative distribution function is given by

$$(e^x - 1)(1 - \cos y)$$

$$1 - e^{-x} \cos y - 1 + \cos y$$

Part C: Written Questions

(10 marks)

Question 1: Consider the following table that presents a random sample of 100 persons selected from the national institute of diabetes. One person is selected at random from the sample

	male	female
Diabetic	38	14
Non-diabetic	22	26

a) what is probability that the person is a female or the person has diabetes?

$$P(F \cup d) = P(F) + P(d) - P(F \cap d)$$

$$= \frac{40}{100} + \frac{52}{100} - \frac{14}{100} = \frac{78}{100}$$

Question 2: Let X be continuous random variable with Probability density function $f(x) = 2ax - 3$, $x \in [1, 3]$ find a .

$$\int_1^3 (2ax - 3) dx = 1$$

$$\left[\frac{2ax^2}{2} - 3x \right]_1^3 = 1$$

$$9a - 9 - [a - 3] = 1$$

$$9a - 9 - a + 3 = 1$$

$$8a - 6 = 1$$

$$8a = 7$$

$$a = \frac{7}{8}$$

Question 3: A problem is given for two students S_1 and S_2 whose chances of solving it correctly are 0.3 and 0.2 respectively. Find the probability that the problem will be solved.

$$P(S_1 \cup S_2) = P(S_1) + P(S_2) - P(S_1 \cap S_2)$$

$$= 0.3 + 0.2 - (0.3)(0.2)$$

$$= 0.5 - 0.06 = 0.44$$